

Python for Artificial Intelligence

Day 3: Loops

Month 1: Python Mastery | LLM & Agentic AI Track

Today's Agenda

- Hour 1: for loops and range(), iterating over strings and lists
- Hour 2: while loops, and loop control: break, continue, pass
- 15-Minute Break
- Hour 3: Problem set, classic interview warm-ups including swapping variables and FizzBuzz

Day 2 Quick Review (10 mins)

Before we start, try these on your own:

1. Write an if/elif/else that grades a score out of 100.

2. What is the difference between a syntax error and a runtime error?

3. What does if my_list: check, without calling len()?

4. Name one common conditional bug.

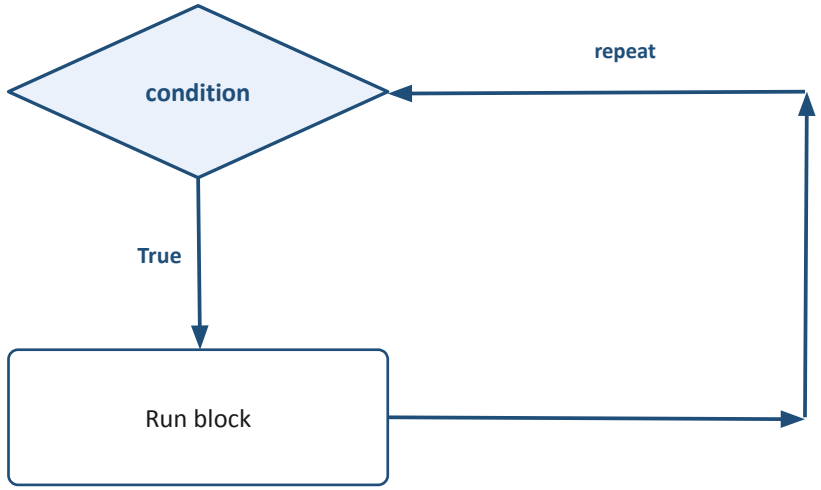
5. You see "KeyError: 'age'" at the bottom of a traceback. What does that tell you?



Hour 1: for Loops

What Is a Loop?

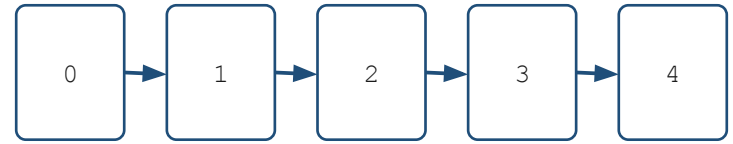
- Repeats a block of code multiple times
- Avoids writing the same code again and again
- Two main types in Python: for and while



False: loop ends, moves to the next line

for Loop & range()

Iterates over a sequence of numbers.



one item per round of the loop

```
for i in range(5):
    print(i)
# Output: 0 1 2 3 4
```

for Loop: Iterating a List

Loops directly over the items of a list.



one item per round of the loop

```
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)
```

for Loop: Iterating a String

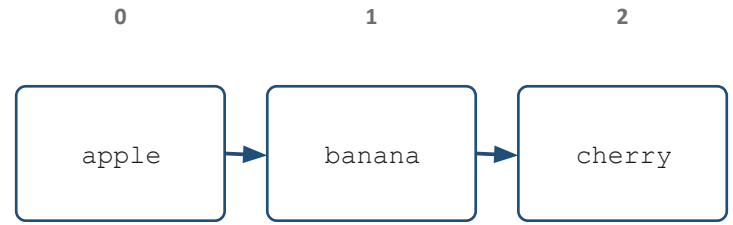
A string is a sequence too, loop over it character by character.

```
word = "python"
for ch in word:
    print(ch)

# common pattern: count something as you go
vowel_count = 0
for ch in "artificial":
    if ch in "aeiou":
        vowel_count += 1
print(vowel_count)
```

for Loop: enumerate()

Get both the index and the value while looping.



index, value pairs, one per round

```
fruits = ["apple", "banana", "cherry"]
for index, fruit in enumerate(fruits):
    print(index, fruit)
```

EXAMPLE 1

Practice: Multiplication Table

Type this code yourself and run it.

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

EXAMPLE 2, INTERVIEW FAVORITE

Practice: Count Vowels in a String

Type this code yourself and run it.

```
text = "Artificial Intelligence"
vowels = "aeiouAEIOU"
count = 0
for ch in text:
    if ch in vowels:
        count += 1
print("Vowel count:", count)
```

QUESTION 1

Think & Practice

Print all even numbers from 1 to 20 using a for loop.

ANSWER

Answer 1

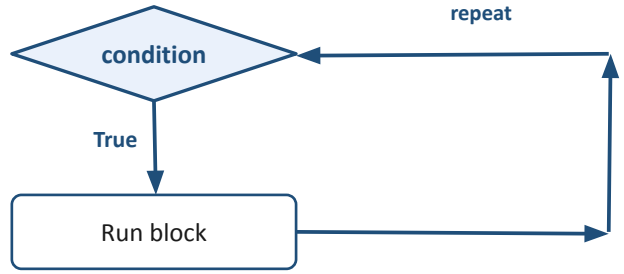
```
for i in range(1, 21):
    if i % 2 == 0:
        print(i)
```



Hour 2: while Loops & Loop Control

while Loop

Repeats as long as a condition stays True.



False: loop ends, moves to the next line

```
count = 0
while count < 5:
    print(count)
    count += 1
```

break, continue & pass

break exits the loop. continue skips to the next round. pass does nothing.



break → exits the loop

continue → skips to next round

pass → does nothing

```
for i in range(10):
    if i == 5:
        break
    if i % 2 == 0:
        continue
    print(i)
```

EXAMPLE 1

Practice: Sum of Natural Numbers

Type this code yourself and run it.

```
n = int(input("Enter n: "))
total = 0
i = 1
while i <= n:
    total += i
    i += 1
print("Sum:", total)
```

EXAMPLE 2, INTERVIEW FAVORITE

Practice: Reverse a String

Type this code yourself and run it.

```
text = "hello"
reversed_text = ""
for ch in text:
    reversed_text = ch + reversed_text
print(reversed_text)
```

EXAMPLE 3

Practice: Guess-Limit Countdown

A while loop that stops itself, common pattern for input validation later.

```
attempts = 0
max_attempts = 3

while attempts < max_attempts:
    guess = input("Enter the secret word: ")
    if guess == "python":
        print("Correct!")
        break
    attempts += 1
    print(f"Wrong. {max_attempts - attempts} attempts left.")
```

QUESTION 1

Think & Practice

Write a while loop that prints the numbers from 10 down to 1, then prints "Liftoff!"

ANSWER

Answer 1

```
n = 10
while n >= 1:
    print(n)
    n -= 1
print("Liftoff!")
```



15-Minute Break

Stretch, grab a coffee. We resume with Hour 3: Interview Problem Set.



Hour 3: Problem Set, Interview Warm-Ups

COMMON INTERVIEW QUESTION

Swapping Two Variables

One of the most frequently asked warm-up interview questions.

```
a, b = 5, 10

# Method 1: using a temporary variable
temp = a
a = b
b = temp

# Method 2: Python tuple unpacking
a, b = b, a

# Method 3: arithmetic (works for integers)
a = a + b
b = a - b
a = a - b
```

QUESTION 1, FIZZBUZZ, VERY COMMON INTERVIEW QUESTION

Think & Practice

Print numbers from 1 to 30.

For multiples of 3, print "Fizz". For multiples of 5, print "Buzz".

For multiples of both 3 and 5, print "FizzBuzz".

ANSWER

Answer 1

```
for i in range(1, 31):
    if i % 15 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

QUESTION 2, HARDEST

Think & Practice (Harder)

Using nested loops, print a right-angled triangle of stars with 5 rows:

```
*

**

***

****

*****
```

ANSWER

Answer 2

```
rows = 5
for i in range(1, rows + 1):
    print("*" * i)
```

Homework: Practice for Interviews

Try these on your own before next class:

1. Write a program that checks whether a number is prime, using a for loop.
2. Write a program that finds the factorial of a number using a loop (not recursion).
3. Write a program that finds the GCD of two numbers using a while loop.
4. Write a program that counts how many times each character appears in a string.
5. Print a number pyramid centered on the screen, for 5 rows.

Key Takeaways

- for loops repeat a known number of times, while loops repeat until a condition becomes False
- break, continue, and pass give direct control over loop flow
- Strings and lists can both be looped over directly, character by character or item by item
- Patterns like FizzBuzz and swapping variables come up often, practicing them now saves time later

Coming Up: Day 4

- Lists & Tuples, In Depth
- Creating, indexing, and slicing lists and tuples
- Mutability and common list methods: append, insert, remove, sort
- Practice set: reverse a list, find duplicates, second-largest element, tuple unpacking